ODWA MTATAMBI

Android Software Engineer

Mobile Engineering • Clean Architecture • AI-Powered Personal Projects

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PROFESSIONAL SUMMARY

Android Software Engineer with 6 years of professional experience delivering production mobile applications across banking and enterprise environments. Skilled in Kotlin, Jetpack Compose, Clean Architecture, MVI/MVVM, and modern Android engineering practices. Outside of core professional work, I explore AI integration and full-stack web development through self-directed personal projects — including a published AI-powered mobile app, an offline-first expense tracker with Firebase and WorkManager sync, and a full-stack portfolio site built with Next.js, React, TypeScript, and Tailwind CSS. Currently deepening expertise through structured self-study, including SSOT with Kotlin Flow, sealed UI-state management, modular DI organisation, and TDD. Writing clean, maintainable code, improving engineering quality through testing and CI/CD, and continuously learning emerging technologies.

KEY ACHIEVEMENTS

- Contributed to feature development on one of South Africa's largest retail banking applications, serving millions of users.
- Built and published Ugrand? (MoodTracker), an AI-powered mental wellness app, to the Google Play Store end-to-end — from architecture and onboarding through a tested database migration and staged release.
- Built and submitted SlipStash to Google Play, an offline-first expense tracker: Room as source of truth, idempotent WorkManager sync, Firebase Auth + Firestore with per-user security rules, and in-app data deletion.
- Built ActionPilotAI, an AI workflow automation platform that turns natural language into executed device actions, open-sourced on GitHub.
- Led an automated UI testing strategy (Appium) covering critical user flows, improving release confidence across teams.
- Designed and deployed a personal portfolio website (Next.js, React, TypeScript, Tailwind CSS) on Vercel as a self-directed project, including custom security headers and CI-driven deployment.
- AWS Cloud Practitioner certified; active technical writer publishing Android architecture and engineering articles on Medium.

PROJECTS

SlipStash — Offline-First Expense Tracker (Submitted to Google Play)

Kotlin · Jetpack Compose · Clean Architecture · MVVM · Hilt · Room · WorkManager · Firebase Auth · Cloud Firestore

- Built an offline-first expense tracker: expenses and receipt photos save to Room instantly, with a live status per expense (Queued, Syncing, Synced, Retry) surfaced through Kotlin Flow, fully usable offline.
- Engineered idempotent background sync with WorkManager — unique work and a unique document ID per expense, skipped re-uploads of synced receipts, network constraints, backoff retries and a periodic sweep — so retries never create duplicates. Verified offline end to end and with a test that syncs one expense twice and asserts one remote document.
- Built a chained receipt pipeline — camera capture via FileProvider, on-device compression with EXIF rotation correction, upload, then expense sync. Receipts sit behind a ReceiptStorage interface, Firestore-backed to stay on the free plan, so moving to Cloud Storage is a single-binding change.
- Integrated Firebase Authentication (anonymous sign-in) and Cloud Firestore, writing per-user security rules so each user can only read and write their own data.
- Built privacy controls: swipe-to-delete with undo, synced to Firestore as an idempotent soft delete, and a "Delete all my data" option that removes cloud data, the anonymous account, the local database and receipt files — cloud first, so a failed delete never orphans data.

- Built the UI in Jetpack Compose (Material 3) with bottom navigation, a monthly summary with category breakdown, and light and dark themes.
- Found a coroutine timeout bug through offline testing — a timeout, a CancellationException subclass, was swallowed and left expenses stuck on Syncing — and fixed it with a regression test. Use cases are unit-tested with JUnit, coroutines-test and hand-written fakes.
- Applied Clean Architecture (repository interfaces, use cases, mappers, Hilt modules including @HiltWorker) and stored money as integer cents to avoid floating-point errors.

NewsOutletApp — Clean Architecture News Reader
Kotlin · Jetpack Compose · Clean Architecture ·MVVM · Hilt ·Retrofit · Paging · DataStore · Coil · Github Actions

- Built a South African news feed app on the NewsData.io REST API with strict Clean Architecture layer boundaries (Presentation/Domain/Data) and interface-based dependency inversion across repositories and use cases.
- Implemented infinite-scroll pagination with Paging 3, backed by a cursor-based PagingSource, with distinct loading, error, and retry states surfaced through sealed UI-state modelling.
- Designed a persistent onboarding flow using Jetpack DataStore (Preferences) with reactive Flow-based state, ensuring returning users skip onboarding without a visible flash on launch.
- Structured dependency injection across dedicated Hilt modules (network, local persistence, repository) with constructor-injected use cases replacing manual object graphs.
- Set up CI/CD with GitHub Actions (build + lint), branch protection requiring checks before merge, and a feature-branch Git workflow.

ActionPilotAI — AI Workflow Automation Platform
Kotlin · Jetpack Compose · MVI · Clean Architecture · Gemini API · Tool Calling · Voice Recognition

- Architected an on-device AI workflow automation platform that converts natural-language voice and text input into real, executed device actions — maps, calendar events, contacts, replies — via structured tool calling against the Gemini API.
- Designed a unidirectional MVI data flow — pure Reducer, sealed Intent classes, one-shot Effects via Kotlin Channel — eliminating impossible UI states and making execution flow fully predictable and testable.
- Built a modular Clean Architecture spanning domain, data, and presentation layers, with a dedicated UseCase layer (GetAiActionUseCase) decoupling orchestration logic from the ViewModel and data sources.
- Automated the delivery pipeline with GitHub Actions and SonarCloud — builds, static analysis, dependency verification, and Jacoco coverage reporting enforced as quality gates on every change.
- Enforced unit test coverage with JUnit and MockK across the ViewModel, UseCase, and Repository layers.

Ugrand? (MoodTracker) — AI-Powered Mental Wellness Application
Kotlin · Jetpack Compose · MVVM · Hilt · Room · DataStore · Coroutines & Flow · Gemini API · Offline-first

- Designed, built and published a mental wellness application to the Google Play Store: users tap a mood, or describe how they feel in their own words, and receive a personalised AI-generated reflection via the Gemini API, informed by their recent mood history.
- Implemented a local-first architecture using Room Database — all user data stored on-device with intelligent offline fallback, ensuring a seamless experience without dependency on external data.
- Reworked the check-in flow so a single tap generates the reading, using coroutine Job cancellation and a log-after-success rule so rapid mood changes never save abandoned entries or display stale messages.
- Built a first-run onboarding flow with HorizontalPager, gated by Preferences DataStore and the SplashScreen API so returning users never see a flash of onboarding.
- Wrote and tested a Room schema migration (v2 → v3) for existing users, verified by upgrading over the published build with mood history preserved.
- Owned the full release pipeline — signing, versioning, R8-enabled App Bundles, internal testing, and Play Console deployment.

WeatherApp — Weather Forecast Android App

Kotlin · Jetpack Compose · Clean Architecture · MVVM · Hilt · Room · Retrofit · Kotlin Flow · CI/CD

- Built a 5-day forecast app on the OpenWeatherMap API with strict Clean Architecture layer boundaries and a Single Source of Truth data pipeline (Room + Kotlin Flow).
- Designed a sealed UI-state hierarchy and composite Room primary keys (date + location) to support multi-city, multi-day data without collision.
- Automated builds and quality checks via GitHub Actions CI/CD, with a network-first cache strategy and graceful offline fallback.

Personal Portfolio Website — https://odwa-portfolio.vercel.app/

TypeScript · React · Next.js · Tailwind CSS · Vercel

- Designed and built a responsive personal portfolio site end-to-end — component architecture, dark/light-mode-aware styling, and accessible navigation.
- Implemented light/dark-aware sections (Hero, Projects, Skills, Highlights, Contact) using reusable, prop-driven React components.
- Configured custom HTTP security headers (X-Frame-Options, X-Content-Type-Options, Referrer-Policy, Permissions-Policy) via Next.js middleware configuration.
- Deployed via Vercel with automatic CI/CD from GitHub, including HTTPS enforcement via HSTS.

PROFESSIONAL EXPERIENCE

Android Software Engineer | Impact.com *Jun 2025 – Feb 2026*

- Led the automated UI testing strategy with Appium, covering critical flows including login, language selection, and theming.
- Delivered the Payout Group feature end-to-end — UI, business logic, and Retrofit API integration.
- Built reusable, accessible Jetpack Compose components adopted across multiple feature teams.
- Mentored teammates on automation setup, architecture, and test-writing; contributed to PR reviews and QA tooling improvements.

Android Software Engineer (Contractor) | Absa Group via LTIMindtree *Jun 2020 – Mar 2024*

- Embedded in the WIMI squad (Wealth Insurance Management) — delivering insurance features within one of South Africa's largest retail banking applications under strict financial compliance constraints.
- Developed multiple customer-facing insurance features end-to-end — freeCover, Pay Arrears and Instant-Life — across the full Android stack from UI through to API integration.
- Refactored insurance features from MVP to MVVM, improving separation of concerns, testability, and long-term maintainability.
- Established unit test coverage across features that previously had none, building a testing foundation that improved release confidence and set a quality standard for the team.
- Integrated RESTful APIs using Retrofit within a heavily regulated banking environment, ensuring implementations met Absa's compliance and security requirements.

Operational Analyst (Contractor) | Absa Group via LTIMindtree *Aug 2019 – Mar 2020*

- Monitored live operational dashboards for a large-scale banking environment, identifying anomalies and service degradations as they surfaced.
- Escalated incidents to the relevant technical and support teams promptly, helping keep disruption to banking services to a minimum.
- Logged and tracked incidents end-to-end in ServiceFirst and later ServiceNow, maintaining accurate records for follow-up and reporting.
- Gained hands-on exposure to ITSM processes and incident management within a regulated financial services setting.

Software Engineer Intern → Software Engineer | LTIMindtree *Feb 2019 – Mar 2025*

- Completed structured training in Java and Android development, progressing from intern to software engineer.
- Built foundational knowledge in Kotlin, Android architecture components, and unit testing practices.
- Earned industry certifications: Brainbench Java 7 and ISTQB Unit Testing.

SKILLS & TECHNOLOGIES

- **Core Stack:** Kotlin, Jetpack Compose, Clean Architecture, MVVM, MVI, Coroutines, Flow, Hilt, Room, Retrofit, Firebase, WorkManager, Azure DevOps, GitHub Actions, Gemini API.
- **Languages:** Kotlin, TypeScript, Java
- **Web & Frontend:** React, Next.js, Tailwind CSS, HTML, CSS
- **Android:** Jetpack Compose, MVVM, MVI, Repository Pattern, UseCase Layer, Hilt, ViewModel, StateFlow, SharedFlow, Kotlin Flow, LiveData, Room Database, Coroutines, Kotlin Channel, WorkManager (chained and periodic work), Preferences DataStore, Navigation Compose, Firebase Authentication, Cloud Firestore, Firestore Security Rules, Material 3
- **Architecture & System Design:** Clean Architecture, SSOT with Flow, Sealed UI States, Composite Room Keys, Modular DI Organisation, Pure Domain Models, Layer Boundary Separation, Offline-First Sync, Idempotent Sync, Soft Deletes, Room Schema Migrations
- **AI Engineering:** Gemini API, AI Workflow Automation, Tool Calling, Prompt Engineering, Natural Language Understanding, Structured Outputs, JSON Parsing Pipelines
- **Networking & APIs:** Retrofit, Ktor, REST APIs, Swagger, OkHttp, Kotlinx Serialization
- **Testing:** Unit Testing, TDD, ViewModel Testing, UI Testing, Appium, MockK, JUnit, Turbine, Jacoco, Code Coverage
- **CI/CD & Quality:** Azure DevOps, GitHub Actions, SonarCloud, Static Code Analysis, Lint, Detekt, Dependency Verification
- **Tools & Deployment:** Git, GitHub, Google Play Store, Vercel, Gradle, R8/ProGuard, Postman
- **Agile:** Scrum, Sprint Planning, PR Reviews, Technical Mentoring
- **IT Service Management:** ServiceNow, ServiceFirst, Incident Management, Incident Logging & Escalation, Operational Dashboard Monitoring

TECHNICAL WRITING

Medium — @odwamtatambi

- Publishes technical articles on Android architecture, software engineering principles, testing, and AI-powered mobile development, focusing on helping developers understand the reasoning behind engineering decisions.

OPEN SOURCE

GitHub — OddzMint-Net

- Open-source Android and AI projects focused on modern architecture, testing, and intelligent workflow automation.

EDUCATION

BSc Computer Science & Geographic Information Systems | University of Fort Hare

CERTIFICATIONS

- AWS Cloud Practitioner
- Generative AI Certification
- Advanced Programming in Kotlin
- Android Studio User Interfaces — Jetpack Compose
- Working with Data on Android (Room, Retrofit, Ktor)
- Version Control (Git, CI/CD, Continuous Deployment)
- Brainbench Java 7 | ISTQB Unit Testing